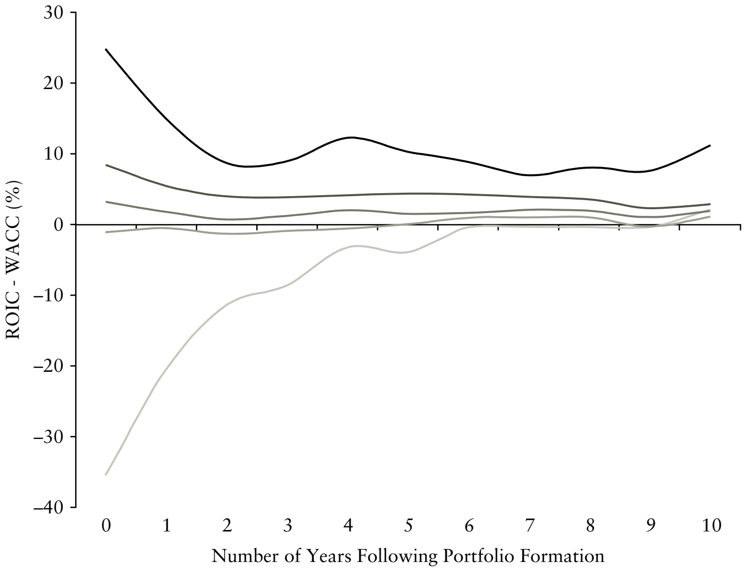

FIGURE 4.3 Change in Median ROIC by Quintile (2000 to 2010)
Source: Michael J. Mauboussin, *The Success Equation: Untangling Skill and Luck in Business, Sports and Investing*. (Boston: Harvard Business Review Press), 2012.

> *So it's not that the standard deviation of the whole sample is shrinking; rather, luck's role diminishes over time.*
>
> *Separating the relative contributions of skill and luck is no easy task. Naturally, sample size is crucial because skill only surfaces with a large number of observations. For example, statistician Jim Albert estimates that a baseball player's batting average over a full season is a fifty-fifty combination between skill and luck. Batting averages for 100 at-bats, in contrast, are 80 percent luck.*

Even so, Mauboussin did find persistence in some businesses' returns on invested capital over the period 2000 to 2010. *Persistence* here is the likelihood a company sustained its return on invested capital over the period examined. He found that a small subset of businesses did generate persistently high returns on invested capital, at levels that could not be attributed solely to chance, but he couldn't identify the underlying causal factors. It's one thing to note *post hoc* that some companies did sustain high or low returns on capital over the period examined, and quite another to do it *ex ante*, or before the fact. The difficulty is that we don't know the factors that *predict* future sustainable high returns on invested capital.